


# Analysis of the anode boundary layer of high intensity arcs

H. A. Dinulescu and E. Pfender

Citation: Journal of Applied Physics **51**, 3149 (1980); doi: 10.1063/1.328063
View online: http://dx.doi.org/10.1063/1.328063
View Table of Contents: http://scitation.aip.org/content/aip/journal/jap/51/6?ver=pdfcov
Published by the AIP Publishing

**Articles you may be interested in**

Measurement of anode falls and anode heat transfer in atmospheric pressure high intensity arcs
J. Appl. Phys. **55**, 714 (1984); 10.1063/1.333129

Analysis of the cathode region of a freeburning high intensity argon arc
J. Appl. Phys. **54**, 3818 (1983); 10.1063/1.332606

Studies of the anode region of a highintensity argon arc
J. Appl. Phys. **53**, 4136 (1982); 10.1063/1.331236

Generation of Ultrafine Aerosols with a Transpiration Cooled Anode in a High Intensity Arc
Rev. Sci. Instrum. **41**, 655 (1970); 10.1063/1.1684610

Heat Intensity and Current Density Distributions at the Anode of High Current, Inert Gas Arcs
J. Appl. Phys. **33**, 1638 (1962); 10.1063/1.1728803




[This article is copyrighted as indicated in the article. Reuse of AIP content is subject to the terms at: http://scitation.aip.org/termsconditions. Downloaded to ] IP: 131.155.81.2 On: Sat, 22 Nov 2014 16:11:36

# Analysis of the anode boundary layer of high intensity arcs

H. A. Dinulescu and E. Pfender
*Heat Transfer Division, Department of Mechanical Engineering, University of Minnesota, Minneapolis, Minnesota 55455*

(Received 29 November 1979; accepted for publication 18 February 1980)

A one-dimensional analysis of the anode boundary layer of an atmospheric pressure, high intensity argon arc reveals substantial deviations from local thermodynamic equilibrium (LTE) in this layer. The temperature of the heavy species approaches the temperature of the anode in the immediate vicinity of the anode surface, whereas the electron temperature remains sufficiently high to ensure the required electrical conductivity. Temperature and density gradients in the anode boundary layer contribute substantially to the electric current flow so that the potential drop across the boundary layer becomes negative. The main voltage drop, which is in the order of 1 V, is essentially confined to the sheath at the bottom of the boundary layer overlying the anode surface. The thickness of this sheath is several orders of magnitude smaller than the anode boundary layer and the potential drop in the sheath is also negative. Therefore, the anode fall becomes negative for the entire parameter range covered in this paper, in contrast to the results of previous theories. This finding affects the anode energy balance as well as the interpretation of indirect (calorimetric) anode fall measurements which is important for the design of arc gas heaters.

PACS numbers: 52.80.Mg, 52.40.Hf

## I. INTRODUCTION

The high pressure, high current (high intensity) arc represents the primary tool for producing thermal plasmas which are of increasing interest for chemical and material processing. In contrast to low intensity or low current arcs, high intensity arcs ($I \geqslant 50$ A) at atmospheric or higher pressures are characterized by strong macroscopic flows induced by the arc itself. Any variation of the current-carrying cross section of the arc leads to induced flows.[1] At sufficiently high currents and axial current density variations, flow velocities in the order of 100 m/s are observed. The cathode jet phenomenon in high intensity arcs is a typical example. In this case, the electron emission process at the cathode causes a contraction of the arc in front of the cathode, which, in turn, gives rise to a continuous magnetohydrodynamic (MHD) pumping action producing the well-known cathode jet. Similar effects may occur in the anode region of high intensity arcs, which is strongly affected by such flows induced by the arc itself.

The anode region may be divided into several zones. A flow-effected zone, as mentioned before, followed towards the anode by a layer in which the presence of the relatively cold anode is felt and which is characterized by steep gradients of temperature and particle densities. In the usual terminology, this layer may be designated as a boundary layer. At the bottom of this boundary layer is the sheath overlying the anode surface.

Over the past years, the flow-effected zone in front of the anode received considerable attention[2,3] in contrast to the anode boundary layer. The strong axial gradients in this layer may cause substantial deviations from local thermodynamic equilibrium (LTE), affecting both chemical as well as kinetic equilibrium.

Although the existence of this layer has been recognized by earlier investigators, a consistent theory is still lacking which would, for example, predict the sign and the magnitude of the anode fall in high intensity arcs. Bez and Hoecker[4] presented an oversimplified picture of the anode fall region with results which could not be confirmed by experiments in the case of thermal arcs. More recently, the anode boundary layer has been studied both experimentally and analytically by a number of investigators from the USSR.[5–8] The experimental work presented in Refs. (5–7) refers essentially to low pressure arcs. In this situation they found that the anode fall is always negative if the arc current surpasses a certain critical value. On the basis of results from probe measurements in arcs at pressures from 5 to 120 Torr, the authors of Ref. 6 concluded that the anode fall should even be negative at atmospheric pressure.

According to the analysis of Ref. 7, the electric field strength in the anode boundary layer should be less than in the adjacent arc column; but the anode fall should still be positive for high intensity arcs. Although there are a number of similarities in the approach and some qualitative agreements with the results of this work, Ref. 7 does not provide sufficient details for a meaningful comparison. In addition, there is a major difference in the choice of the boundary condition for solving the conservation equations for the anode boundary layer between Ref. 7 and this work. This may, at least partially, be responsible for the resulting positive anode falls in Ref. 7.

Unfortunately, there are no pertinent experimental results for a comparison with the predictions in this paper. Previous experimental results (calorimetric measurements of anode losses) have been based on the assumption of positive anode falls. In contrast, the results of this paper show that the anode fall will be, in general, negative in high intensity arcs.

The analysis presented in this paper applies to plasmas composed of atoms, charged positive ions, and electrons. Solutions of the anode boundary layer equations have been ob-

 0021-8979/80/063149-09$01.10 © 1980 American Institute of Physics 

[This article is copyrighted as indicated in the article. Reuse of AIP content is subject to the terms at: http://scitation.aip.org/termsconditions. Downloaded to ] IP: 131.155.81.2 On: Sat, 22 Nov 2014 16:11:36

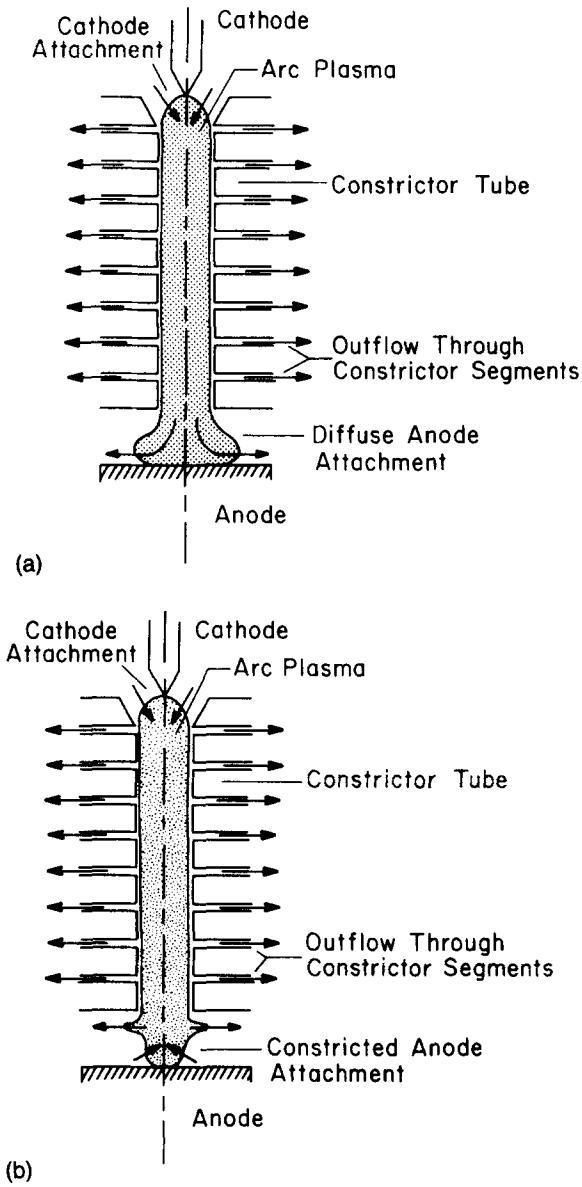



FIG. 1. Cascaded wall stabilized arc. The arrows indicate the local orientation of the mass flow.

tained for arcs in argon at atmospheric pressure and anode current densities ranging from 600 to 3000 A/cm$^2$.

## II. MODELING OF THE ANODE BOUNDARY LAYER

The model adopted for this analysis is based on a rotationally symmetric arc attaching to a plane cooled anode perpendicular to the axis of the arc. Using a cascaded arc arrangement, recent experimental studies in this laboratory[9] revealed two types of anode arc roots as schematically shown in Fig. 1. Figure 1(a) shows a diffuse anode arc root which is caused by strong axial flows from the cathode (cathode jet) and which is classified as "cathode jet dominated mode." Figure 1(b), with a more or less constricted anode arc root due to entrainment of cold gas from the vicinity of the anode, refers to the "anode jet dominated mode." Secondary effects at the anode (melting and/or evaporation) are excluded, i.e., an efficient cooling system shall eliminate such effects.

A blown-up sketch of a constricted arc root is shown in Fig. 2. According to observations of both types of attachments in this laboratory, there is always a relative dark space immediately in front of the anode with a thickness around 0.1–1 mm. This relative dark space seems to be an indication of the anode boundary layer.

Due to its small thickness, the anode boundary layer may be treated as a one-dimensional region. For the purpose of solving the conservation equations, the plasma parameters are assumed to vary only in the direction perpendicular to the anode surface. In an actual anode attachment, there will also be radial variations, in particular if the anode arc root is severely constricted. In the following it will be assumed that the diameter $d$ of an attachment spot at the anode is either large compared to the thickness of the anode boundary layer or, if this is not the case, only the central part of the anode arc root will be considered (small radial gradients). In this way all the radial terms in the conservation equations can be neglected. The radial dependence will enter the problem only through boundary conditions. Since the purpose of this investigation is to obtain general trends within the anode boundary layer, these simplifications are reasonable. In the case of a diffuse anode arc attachment, the previously specified condition $d \gg L$ is always fulfilled.

In Fig. 3, a diagram of the anode boundary layer is presented in which various scale lengths of the plasma are indicated. As will be shown later, the boundary layer thickness is much larger than the particle mean free path length which, in turn, is much larger than the Debye length. Therefore, the ionized gas in the boundary layer may be treated as a continuum and as a true plasma.

## III. THE PLASMA CONSERVATION EQUATIONS IN THE ANODE BOUNDARY LAYER

As pointed out in the previous paragraph, the plasma in the anode boundary layer may be treated as a true continuum. The mass motion, the interdiffusion of the various species, the chemical reactions, and the transport of energy in

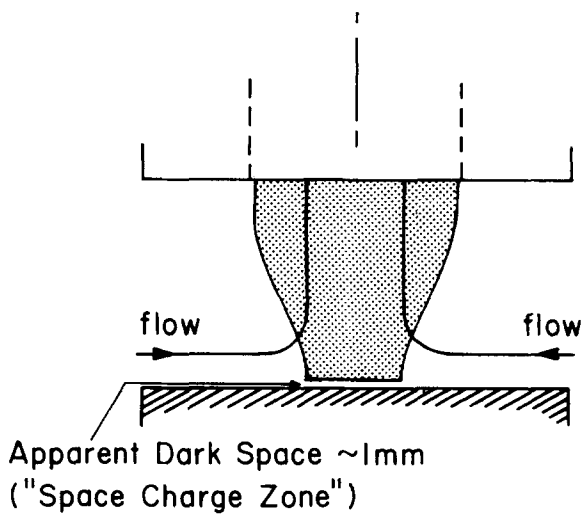



FIG. 2. Constricted anode arc attachment.



[This article is copyrighted as indicated in the article. Reuse of AIP content is subject to the terms at: http://scitation.aip.org/termsconditions. Downloaded to ] IP: 131.155.81.2 On: Sat, 22 Nov 2014 16:11:36

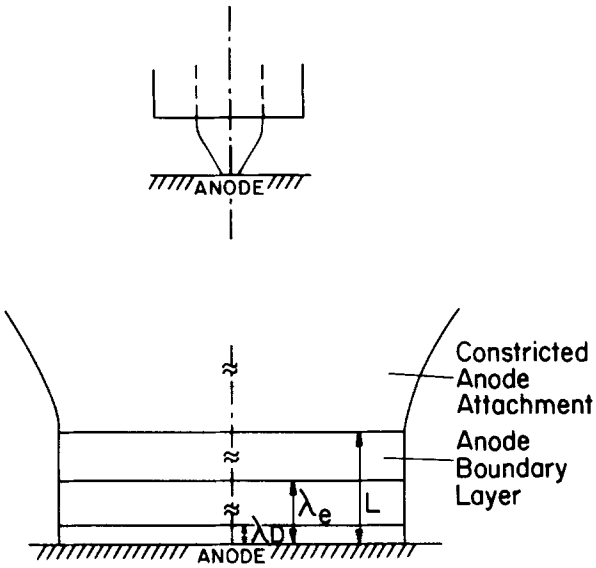



FIG. 3. Scale lengths of plasma in the anode boundary layer. $L$—anode boundary layer thickness; $\lambda_c$—electron mean-free path; $\lambda_D$—Debye radius.

such a plasma are described by conventional equations including the Navier-Stokes equations.

The plasma equations for the anode boundary layer may be derived in different ways. One possibility is to establish volume element balances in which all the relevant forces and fluxes are included. Although this approach enables straightforward physical interpretation, it meets with some difficulties in dealing with several subtle terms (e.g., the thermal diffusion). Another approach makes use of phenomenological expressions for the fluxes of mass, momentum and energy, derived from the theory of irreversible thermodynamics. Although very general, this approach requires that the various transport coefficients (thermal conductivity, electrical conductivity, etc.) are derived from measurements which are extremely difficult due to the high temperature and the very steep gradients. Finally, the most general approach applies solutions of the Boltzmann kinetic equations for the distribution functions of the various species present in the plasma. As soon as the distribution functions are known, the expressions of the various fluxes can be determined, together with the equations for the calculation of transport coefficients in terms of basic physical properties. Any rigorous treatment of macroscopic phenomena in plasmas should be based on Boltzmann's kinetic equations or equations derived from these.

The flux expressions for plasmas treated as continuum have been derived in a number of publications together with procedures for computing the transport coefficients.[10–12] In this work, those expressions have been adopted after carefully checking their validity for the conditions in the anode boundary layer. Since the density of an atmospheric pressure arc plasma is sufficiently high, the heavy components of the plasma (atoms and ions) behave similar to that of an ordinary gas with continuumlike properties as ascertained by the kinetic theory of Chapman and Enskog.[10,11] A special verification of the continuum hypothesis is needed only for the electrons which, due to their charge, are strongly influenced by electromagnetic fields and which, due to their small mass, are weakly bound to the other species of the plasma. It was shown in Ref. 12 that the electron gas can be treated as a continuum using the method of Chapman and Enskog provided that the electron velocity distribution function is quasi-Maxwellian (deviations from the Maxwellian distribution must be of the second order). The conditions for a quasi-Maxwellian distribution of the electron gas are met if there is a high electron-electron collision frequency, a low intensity of the electric fields, and if the plasma is collision dominated. It will be shown that these conditions are indeed fulfilled in the anode boundary layer of a high current-high pressure arc.

If the frequency of the electron-electron collisions is sufficiently high, the directed energy gained by the electrons from the electric field is randomized within the electron gas. Thus, the electrons assume an isotropic velocity distribution. The precise formulation of this condition is

$$\sum_h (m_e/m_h)(\bar{\nu}_{eh}/\bar{\nu}_{ee}) \ll 1 , \tag{1}$$

where the sum extends over all heavy species. The electron mass is denoted by $m_e$, the mass of the heavy particles by $m_h$, and $\bar{\nu}_{eh}$ is the average collision frequency between electrons and heavy particles. Since the electron-ion collision frequency $\bar{\nu}_{ei}$ is of the same order as $\bar{\nu}_{ee}$, the term corresponding to the ion species always satisfies this inequality. The term for the interaction with the neutral species in Eq. (1) establishes a lower limit for the degree of ionization. A numerical estimate for an argon arc plasma may be obtained by using the electron-electron collision cross section computed from the Coulomb potential cutoff at the Debye length, and an energy averaged cross section for the electron-atom collisions of $\bar{Q}_{ea} = 2.\times 10^{-20}$ m$^2$. Assuming an electron temperature of the order of $10^4$ K, one finds that the condition of Eq. (1) is fulfilled if $n_e/n_a \gg 5\times 10^{-9}$, i.e., this condition will always be fulfilled in a high intensity arc, including the anode region.

The condition of small electric field strengths may be expressed in terms of an upper limit for the electron drift velocity[12] namely

$$(U_e/\bar{C}_e)^2 < (m_e/m_h) , \tag{2}$$

where $U_e$ is the electron drift velocity and $\bar{C}_e$ is the average thermal speed of the electrons. A numerical estimate for an argon arc plasma may be based on a current density of $J_e = 10^7$ A/m$^2$, an electron temperature $T_e = 1.2\times 10^4$ K, and an electron number density of $n_e = 6.5\times 10^{22}$ m$^{-3}$ (from Saha equation). By employing the relations

$$J_e = en_eU_e , \quad \bar{C}_e = (8kT_e/\pi m_e)^{1/2} ,$$

one finds that $U_e \approx 9.6\times 10^2$ m/s and $\bar{C}_e \approx 6.8\times 10^5$, which satisfies the condition specified in Eq. (2). The symbol $e$ represents the elementary charge and $k$ represents the Boltzmann constant. Since the parameters chosen for this estimate are representative for a constricted anode attachment, it may be safely assumed that this condition is fulfilled in the anode boundary layer.

Finally, the condition of a collision dominated plasma requires that the local state of the plasma be determined by

v_eh involves v_ei and v_en. v_en is smaller and hence imposes a lower limit on degree of ionizatio n

If electrons are in LTE, then temp of e can be used to determine ne



[This article is copyrighted as indicated in the article. Reuse of AIP content is subject to the terms at: http://scitation.aip.org/termsconditions. Downloaded to ] IP: 131.155.81.2 On: Sat, 22 Nov 2014 16:11:36

collisions of particles with well defined average properties and that radiation to or from the plasma does not affect its state appreciably. This means that collisions should be localized, and that the interactions of particles with the radiation field should be much less probable than the corresponding particle-particle interactions. The condition of localized collisions will be met if macroscopic time and length scales are large compared to the microscopic collision time ($1/\bar{\nu}_{ee}$) and collision free path ($\lambda_e = \bar{C}_e/\bar{\nu}_{ee}$), respectively, i.e.,

$$1/\bar{\nu}_{ee}\tau \ll 1, \quad \text{and} \quad \bar{C}_e/\bar{\nu}_{ee}L \ll 1\,, \tag{3}$$

where $\tau$ and $L$ are the characteristic time and length scales for macroscopic changes. The first part of condition (3) refers essentially to a restriction on the frequency in rf discharges and, therefore, has no significance for dc arcs. In the following, only the second part of condition (3) will be considered. For establishing a more definite expression, $L$ will be taken as the length over which one of the macroscopic properties changes appreciably. Thus with $L \cong T_e/|\nabla T_e|$ and $L \cong n_e/|\nabla n_e|$ conditions (3) may be written as:

$$\lambda_e|\nabla T_e| \ll T_e, \quad \text{and} \quad \lambda_e|\nabla n_e| \ll n_e\,. \tag{3'}$$

For an arc plasma the conditions (3′) are well satisfied within the arc core. Since the values of $\nabla T_e$ and $\nabla n_e$ in the anode boundary layer are not known *a priori*, the condition (3′) shall be assumed to hold in the anode boundary layer. The validity of this assumption will be checked after the solutions of the boundary layer equations are obtained.

The condition of dominant particle collisions over radiation processes may be checked by comparing the probability of collisional excitation and deexcitation with the radiation intensities. If the rates of the various processes are written explicitly, the condition of dominant collisions leads to a lower limit for the electron number density. It was shown in Ref. 12 that for atmospheric pressure argon plasmas, assuming an optically thin situation, the Saha and Boltzmann equilibrium relations may be applied with little error and that the velocity distribution remains quasi-Maxwellian, if the electron number density is maintained above $10^{21}$ m$^{-3}$. From the Saha equation, one finds that this density corresponds to a temperature of approximately 8000 K. The results obtained from the solution of the anode boundary layer equations demonstrate that the electron temperature at the anode remains in excess of 8000 K and the electron number density stays always above the value given by the Saha equation due to diffusion from the hotter parts of the plasma ("frozen" conditions). Therefore, it may be concluded that this condition is also fulfilled, i.e., the plasma in the anode boundary layer is collision dominated.

LTE holds

Individial species are in maxwellian dist

With the conditions for a quasi-Maxwellian distribution fulfilled and assuming in addition that the self magnetic field strength and the macroscopic mass flow are negligible, the electron fluxes are given by the following expressions[12]

$$\mathbf{J}_e = \sigma\left(\mathbf{E} + \frac{1}{en_e}\nabla p_e\right) + \phi\nabla T_e \cong \mathbf{J}\,, \tag{4}$$

$$\mathbf{q}_e = -k\nabla T_e - \left(\frac{5}{2} + \frac{e\,\phi}{k\,\sigma}\right)\frac{kT_e}{e}\mathbf{J}_e\,, \tag{5}$$

where $J_e$, $J$, $\sigma$, $p_e$, $\phi$, and $q_e$ represent the electron current density, the total current density, the electrical conductivity, the partial pressure of the electrons, the thermal diffusion coefficient, and the specific heat flux, respectively.

The heavy particle fluxes can be derived by considering the plasma as a mixture of all its species but ignoring the presence of the electrons. This simplification is justified by the small mass of the electrons and the small energy exchanges between the electron gas and the heavy species.[12] From the theory of Chapman and Enskog[10,11] it follows that the heavy particle fluxes may be written as

$$\mathbf{J}_i = \frac{en_eD_{in}}{kT}\left(\mathbf{E} - \frac{1}{en_i}\nabla p_i\right), \tag{6}$$

$$\mathbf{q}_h = -k_h\nabla T\,, \tag{7}$$

where $J_i$ is the ion current density, $D_{in}$ is the diffusion coefficient of ions among neutral particles, $n_i$ is the ion density, $p_i$ is the partial pressure of the ions, $q_h$ is the specific heat flux due to heavy particles, and $k_h$ is the thermal conductivity of heavy particles.

In addition to the above flux expressions, rate equations are needed for the derivation of the energy equations, namely the rate of ionization, the rate of energy exchange between the electrons and the heavy species, and the rate of radiation escape from the plasma.

The expression of the rate of ionization, taken from Ref. 12, is given by

$$\dot{n}_e = \alpha n_n\left[S(T_e) - (n_e^2/n_n)\right], \tag{8}$$

where $S(T_e)$ is the equilibrium constant in the Saha equation taken at the electron temperature, $\dot{n}_e$ is the rate of ionization, $\alpha$ is the recombination coefficient, and $n_n$ is the density of neutral particles.

The rate of energy exchange in collisions between electrons and heavy particles may be expressed by

V_eh is considerd in ref12 instead of V_ei

$$\dot{E} = (3m_e/m_i)k\,(T_e - T)n_e\bar{\nu}_{ei}\,, \tag{9}$$

where only collisions between the electrons and the ions have been taken into consideration since in the anode boundary layer $\bar{\nu}_{ei} \cong \bar{\nu}_{ee} \gg \bar{\nu}_{en}$, as previously shown.

For the rate of radiative exchange, numerical values may be taken from a series of publications.[13,14] Because of the relatively low temperature levels considered in this paper, the radiation loss term is much smaller than all the other terms in the energy equation and may thus be neglected. Accordingly, for the rate of radiation in the anode boundary layer the approximation

$$\dot{R} = 0 \tag{10}$$

is used.

The various transport coefficients appearing in the previously mentioned flux and rate equations must be computed by using the methods of the kinetic theory as functions of the local plasma properties, i.e., as functions of $n_e$, $n_n$, $T_e$, and $T$. The electron transport properties $\sigma$, $\phi$, $k$, $\alpha$, $\bar{\nu}_{ei}$, have been computed by the methods described in Ref. 15. The heavy particle transport properties $D_{in}$ and $k_h$ have been computed using the method of Chapman and Enskog.[10,11] Procedures for computing transport coefficients as functions of the local plasma properties have been incorporated in the



[This article is copyrighted as indicated in the article. Reuse of AIP content is subject to the terms at: http://scitation.aip.org/termsconditions. Downloaded to ] IP: 131.155.81.2 On: Sat, 22 Nov 2014 16:11:36

computer program for the solution of the plasma transport equations.

With the previously described flux expressions, the conservation equations for the anode boundary layer of an argon arc may be written in one-dimensional form as

$$(*)\quad \frac{d}{dx}\left[\frac{en_eD_{in}}{kT}\left(E-\frac{1}{en_e}\frac{dp_e}{dx}\right)\right]=\dot{n}_e\,, \tag{11}$$

$$J=\text{constant}\,, \tag{12}$$

$$p_i=n_ekT\,, \tag{13}$$

$$\dot{n}_e=\alpha n_n\left(S(T_e)-\frac{n_e^2}{n_n}\right), \tag{14}$$

$$S(T_e)=10.8\times\left(\frac{2\pi m_ekT_e}{h^2}\right)^{3/2}\exp(-\epsilon_i/kT)\,, \tag{15}$$

$$J=\sigma\left(E+\frac{1}{en_e}\frac{dp_e}{dx}\right)+\phi\frac{dT_e}{dx}\,, \tag{16}$$

$$p_e=n_ekT_e\,, \tag{17}$$

$$n_n=\frac{p}{kT}+n_e\left(1+\frac{T_e}{T}\right), \tag{18}$$

$$(*)\quad \frac{d}{dx}\left(k_e\frac{dT_e}{dx}-(\epsilon_i+3.203kT_e)\dot{n}_e\right.$$
$$+3.203\frac{k}{e}J\frac{dT_e}{dx}+JE$$
$$\left.-\frac{3m_e}{m_i}n_e\bar{\nu}_{ei}k(T_e-T)=0\right., \tag{19}$$

$$(*)\quad \frac{d}{dx}\left(k_h\frac{dT}{dx}\right)+\frac{3m_e}{m_i}n_e\bar{\nu}_{ei}k(T_e-T)=0\,, \tag{20}$$

where $x$ is the axial coordinate and $\epsilon_i$ is the ionization energy.

The main unknowns in the set of equations are $n_e$, $T_e$, and $T$. With these key variables all the other quantities of interest can be computed. The equations marked with an asterisk are the main differential equations. The other equations are auxiliary relations.

As previously mentioned, the temperature decreases in the vicinity of the anode from over 10 000 K in the arc plasma to typical anode temperatures of several hundred K. Since the electrical conductivity of an equilibrium argon plasma decreases to almost zero at temperatures below 6000 K, an insulating layer would be formed in front of the anode which makes it difficult to understand how current continuity can be maintained. In the proximity of the anode, however, the plasma is in a state of nonequilibrium in which the species temperature differs appreciably and the gradients of temperature and concentration play an important role for the transport of electric charge carriers.

The solution of the anode boundary layer equations shows that the heavy species temperature decreases to a few hundred degrees in front of the anode whereas the electron temperature remains sufficiently high to maintain an adequate electrical conductivity for the passage of the electric current.

As already mentioned, due to the small thickness of the boundary layer, only variations in the axial direction are important and, therefore, the equations are one-dimensional. At a first glance this may appear as a mere assumption, derived from qualitative experimental observations. One may, however, obtain a realistic estimate of the thickness of the anode boundary layer by computing the axial mean free paths for the pertinent processes. In an argon plasma two processes need to be considered for this purpose: the energy exchange between electrons and heavy particles and the three-body recombination. If the rate of any one of these processes is denoted by $\dot{R}_p$, a characteristic time for the interaction can be defined as

$$\tau_p\equiv n/\dot{R}_p\,, \tag{21}$$

where $\tau_p$ is the characteristic interaction time and $n$ is the number density of the interacting particles. The time $\tau_p$ can be interpreted as the average time interval between two collisions of the interacting particles.

For the process of energy exchange between electrons and heavy particles, a characteristic time is obtained by combining Eq. (9) and (12)

$$\tau_E=\frac{1}{(2m_e/m_i)\bar{\nu}_{ei}}\,, \tag{22}$$

Refering to eq 9, relaxation time is the time required to loose 3/2KT_e of energy

where $\tau_E$ is the characteristic time for energy exchange. It can be interpreted as the average time interval during which the electron loses its excess energy.

For the process of three-body recombination the reaction time is given by

$$\tau_r=n_e/\alpha n_e^2=1/\alpha n_e\,, \tag{23}$$

where $\tau_r$ is the characteristic time for recombination. With the characteristic time known, the axial free path of the reaction can be computed from the relation

$$\lambda_p=U\ \tau_p\,,$$

where $\lambda_p$ is the axial free path of the reaction and $U$ is the drift velocity of the reacting particle. For electrons, the drift velocity is given by

$$U_e=J/en_e\,. \tag{24}$$

Thus, the axial free paths for the processes of interest may be written as

$$\lambda_E=(m_i/2m_e)J/en_e\bar{\nu}_{ei}\,, \tag{25}$$

$$\lambda_r=J/\alpha en_e^2\,, \tag{26}$$

where $\lambda_E$ is the free path length for energy exchange processes, and $\lambda_r$ is the free path length for recombination.

The recombination free path $\lambda_r$ can be interpreted as the distance traveled by an electron in the direction of the electric field between two successive ionization-recombination collisions. The electrons contained in a layer adjacent to the anode of thickness $\lambda_r$ will arrive at the anode without suffering any further ionization-recombination collisions. This means that the charge carrier density in this layer ceases to be controlled by the Saha equation, which assumes that the ionization and recombination processes balance each other perfectly in any volume element. In this layer of plasma, departures from equilibrium are expected to occur and the number densities must be computed from the diffusion equation. Therefore, this layer may be classified as a diffusion boundary layer.



This article is copyrighted as indicated in the article. Reuse of AIP content is subject to the terms at: http://scitation.aip.org/termsconditions. Downloaded to ] IP: 131.155.81.
On: Sat, 22 Nov 2014 16:11:36

TABLE I. Plasma scale lengths in the anode boundary layer.

| $J$ (est.) ($A/m^2$) | $T_e$ (est) (K) | $n_e$ ($m^{-3}$) | $\alpha$ ($m^3/s$) | $\lambda_E$ (mm) | $\lambda_r$ (mm) | $\lambda_e$ (mm) | $\lambda_D$ (mm) |
|---|---|---|---|---|---|---|---|
| $9\times10^6$ | 12 000 | $6.2\times10^{22}$ | $2.6\times10^{-17}$ | 0.2 | 0.6 | $0.2\times10^{-2}$ | $0.3\times10^{-4}$ |
| $1\times10^7$ | 12 500 | $8.1\times10^{22}$ | $2.6\times10^{-17}$ | 0.2 | 0.4 | $0.1\times10^{-2}$ | $0.3\times10^{-4}$ |
| $2\times10^7$ | 13 000 | $1.1\times10^{23}$ | $2.6\times10^{-17}$ | 0.2 | 0.4 | $0.1\times10^{-2}$ | $0.2\times10^{-4}$ |
| $3\times10^7$ | 14 000 | $1.4\times10^{23}$ | $2.7\times10^{-17}$ | 0.2 | 0.3 | $0.1\times10^{-2}$ | $0.2\times10^{-4}$ |

The energy exchange free path $\lambda_E$ can be interpreted as the distance traveled by an electron in the direction of the electric field over which it loses its excess energy by collisions with the heavy particles. This means that in the immediate proximity of the anode the electron temperature is expected to exceed that of the heavy species and the kinetic equilibrium of the plasma ($T_e = T$) breaks down. In the layer of plasma of thickness $\lambda_E$ above the anode, the temperature distribution must be computed by applying a separate energy equation for the electrons and the heavy species. This layer can be classified as a thermal boundary layer. This leads to a refinement of the originally proposed anode boundary layer in the sense that there is a coexistence of a diffusion and a thermal boundary layer. The thickness of the anode boundary layer may be taken as $\lambda_r$ or $\lambda_E$, whichever is larger.

In Table I, estimated values of $\lambda_r$ and $\lambda_E$ are given for a series of current densities and plasma temperatures which have been inferred from existing experimental data. The chosen range for $J$ and $T_e$ pertains to typical values for constricted anode arc roots, while $n_e$ is computed from the Saha equation at the temperature $T_e$. In addition, values of the electron mean-free path for elastic collisions $\lambda_e$ and the Debye length $\lambda_D$ are also listed in Table I.

The values in Table I demonstrate that the thickness of the anode boundary layer ($\lambda_r$ or $\lambda_E$) is indeed small compared to the radius of the anode arc attachment which is typically in the order of several millimeters; but, at the same time this layer is much larger than the electron mean-free path, which justifies the continuum approach. Since the Debye length is much smaller than the thickness of the anode boundary layer and even smaller than the electron mean-free path, the ionized gas in the boundary layer may be considered as a true plasma (the condition of quasineutrality is fulfilled) and the electrical sheath in front of the anode may be included in the analysis as a boundary condition. It should be emphasized that the values listed in Table I are only estimates; the various scale lengths can be accurately computed only after the solution of the complete set of equations has been obtained. The final results show a reasonable agreement between estimated and exact values.

## IV. SOLUTIONS OF THE ANODE BOUNDARY LAYER EQUATIONS

In order to solve the system of equations, appropriate boundary conditions must be specified. These boundary conditions are indicated in Fig. 4.

At the interface between arc column and anode boundary layer: (1) The heavy particle temperature, $T = T_0$, and the electron temperature, $T_e = T_{eo} \cong T_0$; spectrometric measurements in this laboratory[16] confirm the validity of this boundary condition, (2) The heavy particle temperature gradient $dT/dx = (dT/dx)_0$ and the electron temperature gradient $dT_e/dx = (dT_e/dx)_0 = (dT/dx)_0$ are identical, and (3) The charge carrier number density $n_e = n_{eo}$ and its gradient $dn_e/dx = (dn_e/dx)_0$, are both computed from the Saha equation at $T_{eo}$ and from $(dT_e/dx)$. At the interface of the sheath and the anode boundary layer, the heavy particle temperature is equal to that of the anode surface, $T = T_w$.

The system of Eqs. (11)–(20) is essentially a system of three second-order differential equations, which requires for its solution six boundary conditions. Seven boundary conditions have been specified because the thickness of the anode boundary layer $L$ is an additional unknown in this problem. Although the solution of the system of boundary layer equations is completely determined by the seven boundary conditions, the sheath overlying the anode surface requires additional attention. The extremely small thickness of the sheath ($\lambda_D \ll \lambda_e$) requires only that its contribution to the potential drop is known. It was found in this work that the sheath potential drop is negative for all the cases investigated. The potential drop in the sheath may be determined by comparing the total electron current arriving at the anode with the random current in the vicinity of the anode, i.e.,

$$\phi_0 = -(kT_e/e)\ln(en_{ea}\bar{C}_e/4J),$$

where $n_{ea}$ is the electron number density at the interface of the sheath and the boundary layer, and $\phi_0$ is the potential drop across the sheath. The sign of the potential drop across

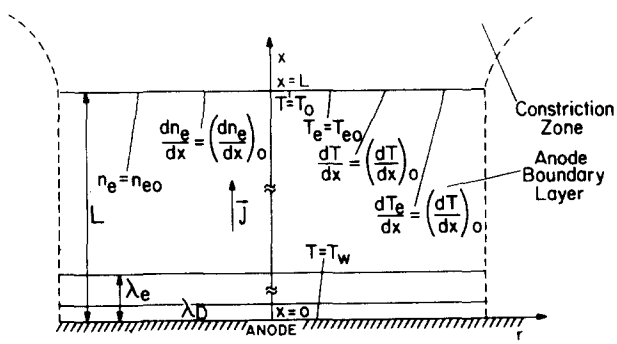



FIG. 4. Boundary conditions for the anode boundary layer. $n_e$—charge carrier number density; $T_e$—electron temperature; $T$—heavy particle temperature; $T_w$—wall temperature; $J$—electric current density; $\lambda_e$—electron mean-free path; $\lambda_D$—Debye length; $L$—anode boundary layer thickness.



[This article is copyrighted as indicated in the article. Reuse of AIP content is subject to the terms at: http://scitation.aip.org/termsconditions. Downloaded to ] IP: 131.155.81.2 On: Sat, 22 Nov 2014 16:11:36

the sheath is determined by the relative magnitude of the actual current density $J$ and the random current density $en_{ea}\frac{1}{4}\bar{C}_e$ at the sheath edge which may be larger or smaller than $J$. For $en_{ea}\frac{1}{4}\bar{C}_e > J$ the potential drop will be negative and for $en_{ea}\frac{1}{4}\bar{C}_e < J$ the potential drop will be positive. The actually prevailing conditions cannot be specified *a priori*, i.e., $n_{ea}$ and $\bar{C}_e = f(T_e)$ must be found from solutions of the anode boundary layer equations. It is obvious that high current densities $J$ in combination with relatively low values of $T_e$ and/or $n_{ea}$ may produce positive anode falls.

The system of equations with the specified boundary conditions is solved by using a numerical method. Among several available methods, the Runge-Kutta procedure appears to be advantageous because of the particular manner in which the boundary conditions are specified. The boundary conditions at the interface between arc column and anode boundary layer constitute a complete set of initial conditions for the system of equations. By starting from this outer boundary and marching toward the anode, the supplementary condition $T = T_w$ may be recovered. At this point the anode surface is assumed to have been reached and the length covered by the marching process is taken to be equal to the anode boundary layer thickness $L$. The described procedure allows the determination of the boundary layer thickness in addition to the local values of the parameters $n_e$, $T_e$, and $T$. This procedure is only feasible because the axial coordinate $x$ does not appear explicitly in the system of equations.

The previously described system of equations is solved for an atmospheric pressure argon arc for five different values of the anode current density ranging from 600 A/cm$^2$ to 3000 A/cm$^2$.

## V. RESULTS AND DISCUSSION

Figures 5–10 show a typical set of solutions for an anode current density of 900 A/cm$^2$. The temperature field is presented in Fig. 5. Values of temperature and temperature gradient at the interface between LTE plasma and anode boundary layer are specified as follows:

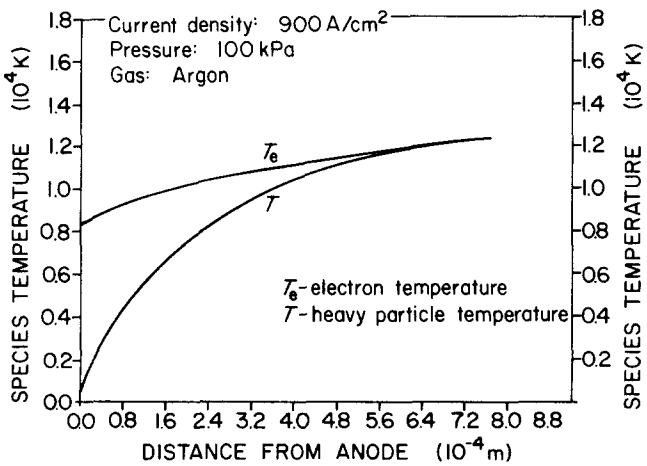



FIG. 5. Species temperature distribution in the anode boundary layer (SI units).

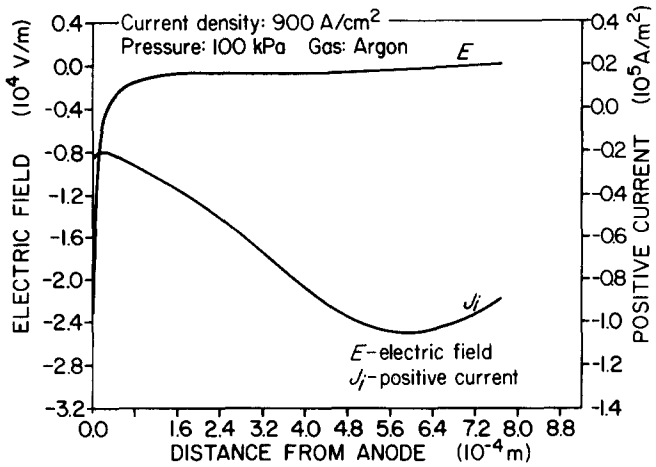



FIG. 6. Electric field and positive current distribution in the anode boundary layer (SI units).

$$T_0 = T_{e0} = 12\,400 \text{ K},$$

$$\left(\frac{dT}{dx}\right)_0 = \frac{dT_e}{dx} = 2.754\times 10^6 \text{ K/m}.$$

These boundary conditions have been selected as representative from a series of spectrometric measurements in the anode region of high intensity arcs.[16] The heavy particle temperature decreases from 12 400 K in the plasma to 400 K which is assumed to be the temperature of the anode surface. The electron temperature decreases much slower, from its value in the plasma of 12 400 K to 8430 K at the anode surface. The two temperature character of the plasma in the proximity of the anode surface is quite pronounced.

The electron temperature maintains sufficiently high values over the entire thickness of the anode boundary layer to ensure the required electrical conductivity for the passage of the electric current.

Figure 6 shows local values of the electric field $E$ and of the positive current density $J_i$. The electric field, starting from a small positive value of 3.5 V/m at the arc column side, turns negative inside the boundary layer and assumes large negative values up to $-2.8\times 10^5$ V/m in the vicinity

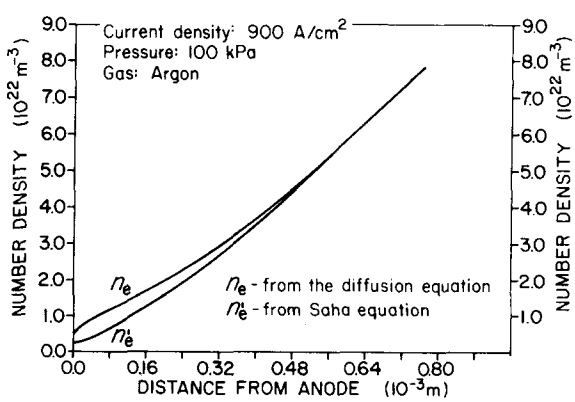



FIG. 7. Charge charrier number density distribution in the anode boundary layer (SI units).



[This article is copyrighted as indicated in the article. Reuse of AIP content is subject to the terms at: http://scitation.aip.org/termsconditions. Downloaded to ] IP: 131.155.81.2 On: Sat, 22 Nov 2014 16:11:36

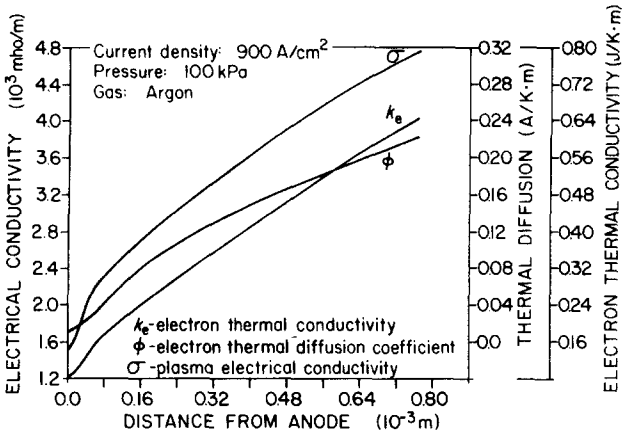



FIG. 8. Electron transport properties in the anode boundary layer (SI units).

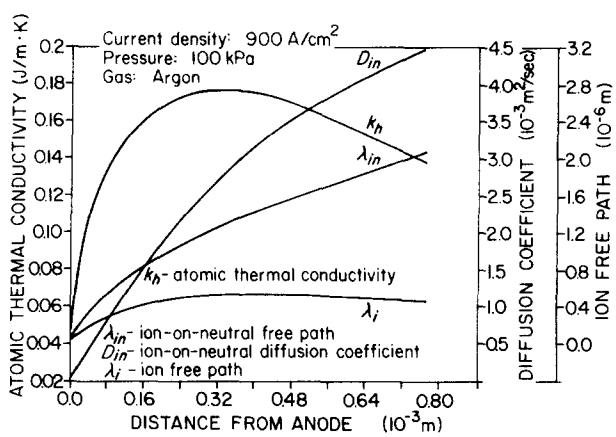



FIG. 10. Heavy particle transport properties in the anode boundary layer (SI units).

of the anode indicating the presence of the electrical sheath. The sheath itself does not appear in the plot because of its small thickness. For this case, the computed value of the potential drop in the electrical sheath is $\Phi_0 = -1.8$ V. This is much larger than the potential drop obtained by integrating the electrical field strength over the entire thickness of the anode boundary layer, i.e., the anode fall is essentially confined to the sheath. The negative value of the potential drop signifies that the electric field in the anode boundary layer is opposing the motion of the electrons and accelerating positive ions. The positive ion current density $J_i$ is directed toward the anode over the entire thickness of the anode boundary layer. The ions traveling in the same direction as the electrons are neutralized as they reach the anode surface which is assumed to be perfectly "catalytic" (no charged particles are reflected from the anode). As the ions approach the anode, the ion current density decreases due to the increase of the neutral particle number density which reduces the ion mobility. Far away from the anode, the ion current will be positive (directed away from the anode). Figure 7 shows the distribution of the electron density within the boundary layer. Besides the actual value $n_e$ computed from the diffusion Eq. (11), the equilibrium (Saha) number density $n_e'$ is plotted for comparison. In the proximity of the anode surface, the actual value is about twice as large as the Saha value. This is due to the fast diffusion of the electrons toward the anode. Since there is not enough time for the electrons to recombine before reaching the anode, substantial deviations from chemical equilibrium occur ("frozen" condition).

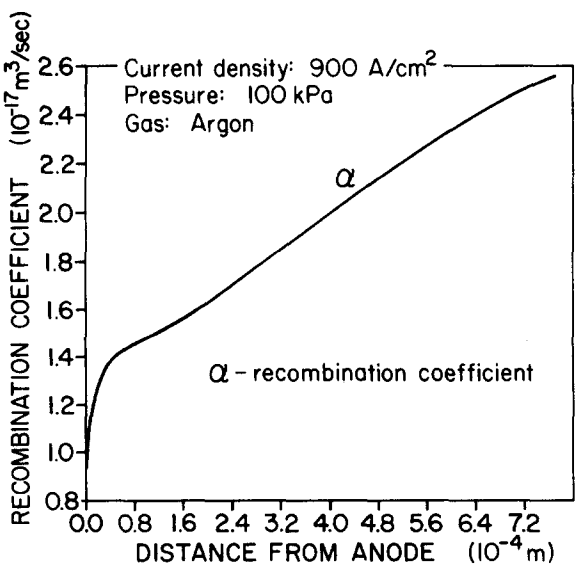



FIG. 9. Recombination coefficient in the anode boundary layer (SI units).

Figures 8–10 present the local values of the transport properties, which are employed for the solution of the differential equations. The transport properties are, in general, functions of $T_e$, $T$, $n_e$, and $n_n$ which is taken into account in computing local values of those coefficients.

A summary of the results for all five current densities is presented in Table II where the thickness of the anode boundary layer $L$ is defined arbitrarily as the distance from the anode to the point at which the difference $T_e - T$ becomes less than 10% of the electron temperature $T_e$. The asymptotic approach of the electron temperature to the temperature of the heavy particles with increasing distance from the anode requires a cutoff at a certain (arbitrary) location. This procedure facilitates comparisons for different anode current densities. The results indicate that the thickness of the anode boundary layer decreases as the current density increases. The potential drop across the boundary layer (the anode drop) which includes the potential drop in the sheath

TABLE II. The influence of the current density on anode boundary layer thickness and potential drop across the sheath.

| $J$ (A/cm²) | $T$ (K) | $L$ (mm) | $\Phi_0$ (V/m) |
|---|---|---|---|
| 600 | 10 000 | 0.53 | −2.2 |
| 900 | 12 400 | 0.34 | −1.8 |
| 1000 | 12 000 | 0.32 | −1.9 |
| 2000 | 13 000 | 0.20 | −1.5 |
| 3000 | 14 000 | 0.12 | −2.7 |



[This article is copyrighted as indicated in the article. Reuse of AIP content is subject to the terms at: http://scitation.aip.org/termsconditions. Downloaded to ] IP: 131.155.81.2 On: Sat, 22 Nov 2014 16:11:36

is negative in all the cases investigated. There appears to be no general trend of this potential drop as a function of the current density which is not surprising because the computed value of $\Phi_0$ (as well as that of $L$) depends also on the plasma temperature. More extensive calculations are needed in order to draw a final conclusion about the trend of these parameters.

Finally, the expression for the heat flux at the anode surface may be written as

$$|q|_{x=0} = k_e \left| \frac{dT_e}{dx} \right| + k_h \left| \frac{dT}{dx} \right| + J[3.203(kT_e/e) + W],$$

where $W$ is the work function of the anode material. The anode potential drop does not appear in this equation. This term appears only if the anode drop is assumed to be positive. In previous work (see for example Ref. 17) anode falls were assumed to be positive in high pressure–high current arcs. For the conditions of the present investigation, the contribution of the ion current to the anode heat flux is negligibly small and has been accordingly ignored.

## ACKNOWLEDGMENTS

This work has been supported by the National Science Foundation under grant NSF/ENG-7704108.

[1]H. Maecker, Z. Phys. **141**, 198 (1955).
[2]E. Pfender and J. Schafer, J. Heat Transfer C **97**, 1, 41 (1975).
[3]E. Pfender, in *Gaseous Electronics* (Academic, New York, 1978, Chap. 5.
[4]W. Bez and K. H. Höcker, Z. Naturforsch **9** Teil A, 72–81 (1954); **10** Teil A, 714–717 (1955); **11** Teil A, 118–123, 192–196 (1956).
[5]G. A. Dyuzhev, S. H. Shkol'nik, and V. G. Yur'ev, Zh. Tekh. Fiz. **48**, 1195–1312 (1978).
[6]N. S. Merinov and V. A. Petrosov, Zh. Prikl. Mekh. Tekh. Fiz. No. 1, 17–24 (1976).
[7]V. A. Nemchinsky and L. N. Peretts, Zh. Tekh. Fiz. **47**, 1868–1875 (1977).
[8]N. S. Merinov, I. N. Ostretsov, V. A. Petrosov, and A. A. Porotnikov, Zh. Tekh. Fiz. **46**, 806–813 (1976).
[9]N. A. Sanders, M. Sc. thesis, University of Minnesota, 1979.
[10]S. Chapman and T. G. Cowling, *The Mathematical Theory of Non-Uniform Gases*, 3rd ed. (Cambridge U. P., Cambridge, 1970).
[11]J. O. Hirschfelder, C. F. Curtiss, and R. B. Bird, *Molecular Theory of Gases and Liquids* (Wiley, New York, 1964).
[12]M. Mitchner and C. H. Kruger, Jr., *Partially Ionized Gases* (Wiley, New York, 1973).
[13]H. W. Emmons, Phys. Fluids **10**, 1125–1136 (1967).
[14]D. L. Evans and R. Tankin, Phys. Fluids **10**, 1137–1144 (1967).
[15]H. A. Dinulescu, Ph. D. thesis, University of Minnesota, 1979.
[16]K. Etemadi (unpublished).
[17]E. R. G. Eckert and E. Pfender, in *Advances in Heat Transfer* (Academic, New York, 1967), Chap. 4.



[This article is copyrighted as indicated in the article. Reuse of AIP content is subject to the terms at: http://scitation.aip.org/termsconditions. Downloaded to ] IP: 131.155.81.2 On: Sat, 22 Nov 2014 16:11:36